Mayu Tatsumi

mtatsumi@g.hmc.edu

linkedin.com/in/mayutatsumi | github.com/mayu-tatsu | +1 (929) 468-0988

EDUCATION

Harvey Mudd College, *Claremont, CA.* 3.75 GPA; B.S., Electrical Engineering; 2027

Relevant Coursework: Microprocessor Design, Advanced Systems Engineering II, RF Circuit Design, Digital Electronics & Computer Architecture, Data Structures

Great Neck South High School, *Great Neck, NY.* 4.0 GPA; 2023

SKILLS

Technical Skills: SystemVerilog, C, Python, MATLAB Simulink, Linux, LTSpice, Solidworks CAD, Excel & Word

Languages: English (Native), Japanese (Native), Spanish (Basic), Korean (Basic)

RELEVANT EXPERIENCE

Fluorescence-Guided Surgery Camera System; PATH Imaging *August 2026 ~ Present*

- Designed camera sensor control, calibration, and image processing algorithms for color and fluorescence camera sensors.

SW Engineering Intern; Yaskawa Solectria Solar *May ~ August 2026*

- Modeled, extracted, and compared frequency response of MATLAB Simulink model and physical lab PV inverter utilizing small-signal perturbation and control systems.
- Audited and began migrating a 60+ package Jenkins/Debian build pipeline from 32-bit to 64-bit NXP i.MX ARM chip; converted C binaries to portable shell scripts and validated changes against live hardware.

Data Acquisition System; Indomo *January ~ May 2026*

- Designed signal filters and wrote Arduino code to interface with 9-DoF IMU and provide real-time calculated shake intensity and rotation over time in MATLAB.

Next-Gen Innovators Mentorship Programme; Apple *September 2025 ~ May 2026*

Control Systems Intern; TechMagic Inc. *May ~ June 2025*

- Designed, implemented, and debugged C firmware for an NXP i.MX8M series processor, enabling a nationwide firmware update for deployed robots.
- Designed tests and edge cases to verify firmware reliability outlined in design reviews.
- Side Projects: Remaking GUI in Flutter, creating MQTT modules, and reconfiguring the Watchdog system.

PREVIOUS ROLES

Grader & Tutor; Harvey Mudd College *August 2024 ~ Present*

Courses: E101: Advanced Systems Engineering I, E72: Applied Math for Engineering, E79: Engineering Systems, CS42: Principles and Practice

MuddLife Co-Director; Harvey Mudd College *August 2025 ~ May 2026*

- Executive board constituency of the Associated Students of Harvey Mudd College (ASHMC).
- Organized and hosted on-campus events, six events per semester, for the rest of the college.

Software Engineering Internship; TVision Insights *May ~ August 2024*

- Optimized a YOLOv7 CNN human recognition model with Intel's OpenVINO toolkit for deployment on Intel NUCs.
- Converted a TensorFlow model to PyTorch and integrated it into the system architecture.

PROJECTS

Karaoke Machine; Final project for µP course *November ~ December 2025*

- Built a karaoke machine that plays a song, displays lyrics, detects the singer's notes, and computes a grade at the end.
- Designed filter stages and coefficients for CIC, FIR, and half-band filter, implemented them on FPGA (SystemVerilog).
- Developed custom DMA and SPI peripheral drivers on an STM microcontroller, utilizing interrupts (C).
- Interfaced with LCD, speaker, and PDM MEMS digital microphone hardware components.